Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

| Proposer | Name | Sangkyun Cho, PhD | Email | tipa@jhu.edu |
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## 2. Technology Overview

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| Title of Proposed Technology | High-throughput, Standardized Cardiac Organoid-Based Toxicity Evaluation Array and Analysis Platform | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | Bio Medical | | Bio Medical | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | Bio-processes / Bio-Equipment | Bio-process Technologies | Bio-processes / Bio-Equipment | Bio-Analytical Instruments |
| Concept of Proposed Technology | The proposed technology integrates Johns Hopkins University's high-functionality cardiac organoid culture technology with Korean SME's automated mass production platform to develop a well-standardized, high-throughput cardiac organoid tissue array for cardiotoxicity screening.<br><br>To overcome the limitations of animal models and 2D cell cultures in predicting human drug responses, this project establishes a system for the homogeneous mass production (CV < 15%) of human iPSC-derived cardiac organoids. These are transformed into an assay-ready platform optimized for high-throughput toxicity screening and global CRO services. | | | |
| Rationale for Joint R&D Project | 1. **The Problem**: Cardiotoxicity (e.g., arrhythmia) is a primary cause of drug attrition and market withdrawal. However, current cardiac organoid systems, despite their significant potential as scalable and physiologically relevent human tissue models, suffer from significant batch-to-batch heterogeneity and high production costs, making them unsuitable for commercial toxicity testing<br><br>2. **The Solution**: By combining JHU's "biomimetic differentiation protocol" with Korean SME's "scaffold-free automated culture & sorting technology," we create a "Hardware-Software" ecosystem. This enables the production of thousands of structurally and functionally uniform cardiac organoids | | | |

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| | per batch, loaded onto standardized test plates.<br><br>3. **Why Joint R&D?**: Johns Hopkins contributes their domain expertise in cardiac organoid differentiation and characterization protocols, and provides the essential multi-omics analysis and validation metrics for clinical relevance. The Korean SME handles the manufacturing standardization, workflow streamlining, and commercialization. This division of labor ensures the technology is both scientifically rigorous and commercially viable. |
| **Key Words** | **1. Cardiac Organoid**<br><br>**2. Cardiotoxicity Screening**<br><br>**3. Standardized Mass Production**<br><br>**4. Multi-omics Analysis**<br><br>**5. Drug Safety Assay** |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level (TRL 3–4) | End of the Project | Level (TRL 5–6) |
|---|---|---|---|---|
| R&D Objectives | Establishment of a Mass Production Process for Cardiac Organoids and Commercialization of a High-throughput Toxicity Evaluation Array<br>- Secure yield and homogeneity for iPSC-derived cardiac organoids<br>- Develop cardiotoxicity evaluation indices compliant with global guidelines (ICH)<br>- Manufacture toxicity assay kits and launch CRO services | | | |
| R&D Target Specifications | 1. Organoid Production<br>• Production scale: ≥ 1,000 organoids per batch<br>• Size/Morphology uniformity: target diameter = 1.2 mm (CV ≤ 15%); circularity > 0.85)<br>• Beating uniformity: target beating rate = 55 BPM (CV ≤ 15%)<br><br>2. Biological Functionality<br>• Maturity marker expression (cTnT, TNNT2): ≥ 80% similarity to normalized expression levels and coverage/distribution in human heart tissue<br>• Contractility: fractional area change (FAC) > 8% (CV ≤ 20%); edge displacement > 2% (CV ≤ 20%)<br>• Calcium handling (Fluo-4): transient amplitude (ΔF/F0) ≥ 0.4; time to peak (TTP) 40-180 ms; decay time (tau) 100–250 ms<br><br>3. Assay Performance<br>• Toxicity prediction accuracy: ≥ 90% (Sensitivity/Specificity)<br>• Z-factor: ≥ 0.5 (Suitable for high-throughput screening)<br><br>4. Commercialization<br>• MoU with Global CRO service providers<br>• PoC with global pharmaceutical companies for drug toxicity and efficacy screening | | | |

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| **R&D Plan** | **Phase 1: Foundation & Optimization**<br>• (JHU) Optimization of differentiation protocols for high-function cardiac organoids.<br>• (Korean SME) Setup of initial mass production using automated culture systems. Optimization of organoid array for compatibility with multi-omics analyses.<br>• (Joint) Establishment of Quality Control (QC) criteria.<br><br>**Phase 2: Scale-up & Data Accumulation**<br>• (JHU) Functional characterization followed by multi-omics (transcriptomics/proteomics) analysis using standard toxicity drug libraries.<br>• (Korean SME) Refinement of AI-based sorting algorithms to minimize batch-to-batch variation (CV < 15%).<br>• (Joint) Construction of drug response database and development of toxicity scoring algorithms.<br><br>**Phase 3: Commercialization & Service Launch**<br>• (JHU) Blind testing to validate the platform's predictive accuracy.<br>• (Korean SME) Fabrication of prototype toxicity assay kits and launch of pilot CRO services.<br>• (Joint) SOP standardization and global market promotion. |

## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | [Bio-Validation & Analytics]<br>• Protocol Development: Provide proprietary protocols for mature cardiac organoid differentiation.<br>• Multi-omics Analysis: Conduct multi-omics analysis (10X Genomics Visium HD) to elucidate/confirm drug Mechanisms of Action (MOA) and to evaluate drug toxicity across an array of multiple organoids simultaneously.<br>• Index Development: Develop quantitative biomarkers and indices for cardiotoxicity (arrhythmia, cytotoxicity).<br>• Academic Validation: Publish findings in high-impact journals to establish scientific credibility. |
| **Expected R&R of Korean SME Partner** | [Manufacturing & Commercialization]<br>• Mass Production: Establish standardized manufacturing processes using 3D floating culture and AI sorting.<br>• Hardware Development: Design and manufacture assay plates optimized for toxicity screening and multi-omics analysis.<br>• QC System: Implement industrial-grade QC SOPs to ensure product consistency.<br>• Global Commercialization: Launch CRO services and assay kits for global pharmaceutical companies. |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | [Scientific Outcomes]<br>• Establishment of a human-relevant cardiac organoid array platform to replace animal testing.<br>• Deepened understanding of cardiotoxicity mechanisms via multi-omics analysis.<br><br>[Technological & Industrial Impact]<br>• Cost Reduction: Improve drug development success rates by early detection and evaluation of cardiotoxicity<br>.<br>• Standardization: Set a global standard for organoid homogeneity and mass production technology.<br><br>[Commercial & Societal Impact]<br>• Market Entry: Enter the high-value global CRO market with proprietary toxicity screening services.<br>• Global Collaboration: Strengthen the credibility of Korean bio-ventures through partnership with a world-leading institute. |

## 6. References


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